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Appendix

A Dataset Construction

A.1 Activity Scheduling

Problem model. We work on the standard unweighted interval scheduling problem. Each instance has m activities indexed by $I = \{1, \ldots, m\}$ with start/finish times (s_i, f_i) and $s_i < f_i$. Intervals are treated as *half-open* $[s_i, f_i)$, so touching endpoints are compatible ($f_j \leq s_i$ means i and j can both be selected). We fix a deterministic tie-breaking order by the tuple (f_i, s_i, i) whenever sorting is required. The canonical reporting order for ground truth is by non-decreasing f (ties by smaller i).

Sampling on a minute grid. All times lie on an integer minute grid. To construct a candidate set of m activities, we: (i) sample $m \sim \mathrm{Unif}\{m_{\min}, \ldots, m_{\max}\}$; (ii) repeatedly sample integer start times $s \sim \mathrm{Unif}\{0, \ldots, S_{\max}\}$ and integer durations $d \sim \mathrm{Unif}\{D_{\min}, \ldots, D_{\max}\}$, set $f = s + d$, and accept the interval if $f \leq S_{\max} + D_{\max}$; we continue until m intervals are accepted. (Values used in our experiments: $m_{\min} = 5$, $m_{\max} = 16$, $S_{\max} = 9 \times 60$, $D_{\min} = 10$, $D_{\max} = 120$; any fixed choices are acceptable as long as $D_{\min} \geq 1$ and times are integral.) Each accepted interval receives a stable ID $i \in \{1, \ldots, m\}$ in order of creation; the table shown to the model lists rows by ID with times in `HH:MM`.

Uniqueness check by DP counting. Write $J = (1, \ldots, n)$ for the intervals *sorted* by (f_i, s_i, i) (we reuse the symbol i for the sorted index when clear). Define the predecessor map

$$p(i) \;=\; \max\{\, j < i \;:\; f_j \leq s_i \,\} \quad \text{with } p(i) = 0 \text{ if the set is empty.}$$

Let $\mathrm{opt}[i]$ be the maximum feasible cardinality using only $\{1, \ldots, i\}$ and let $\mathrm{cnt}[i]$ be the *number of distinct schedules* that attain $\mathrm{opt}[i]$ using $\{1, \ldots, i\}$. Initialize $\mathrm{opt}[0] = 0$ and $\mathrm{cnt}[0] = 1$ (one empty schedule). For $i = 1, \ldots, n$ set

$$\begin{aligned}
\textsc{incl} &= 1 + \mathrm{opt}[\,p(i)\,], \\
\textsc{excl} &= \mathrm{opt}[\,i-1\,], \\
\mathrm{opt}[i] &= \max\{\textsc{incl}, \textsc{excl}\}, \\
\mathrm{cnt}[i] &= \begin{cases} \mathrm{cnt}[\,p(i)\,], & \text{if } \textsc{incl} > \textsc{excl}, \\ \mathrm{cnt}[\,i-1\,], & \text{if } \textsc{excl} > \textsc{incl}, \\ \mathrm{cnt}[\,p(i)\,] + \mathrm{cnt}[\,i-1\,], & \text{if } \textsc{incl} = \textsc{excl}. \end{cases}
\end{aligned}$$

Then $k^\star = \mathrm{opt}[n]$ is the optimal size and $\mathrm{cnt}[n]$ is the number of optimal schedules. If $\mathrm{cnt}[n] = 1$, the optimum is unique. Under uniqueness we reconstruct the unique optimal set $S^\star$ by backtracking from $i = n$: when $\textsc{incl} > \textsc{excl}$ we *include* i and jump to $p(i)$; when $\textsc{excl} > \textsc{incl}$ we *exclude* i and go to $i - 1$; when equal, exactly one branch leads to a nonzero count under uniqueness—choose the branch whose downstream count equals 1 (include if $\mathrm{cnt}[p(i)] = 1$ and $\mathrm{cnt}[i-1] = 0$, otherwise exclude).

Greedy sanity check and canonicalization. Because earliest-finish-time is optimal for unweighted interval scheduling, if the optimum is unique then the greedy policy with the same tie-breaks must return that same set. We nevertheless perform a *sanity check*: run a single pass of earliest-finish-time using the order (f_i, s_i, i) to obtain G; we accept the instance only if $G = S^\star$ (as sets). The ground-truth sequence we release is the unique set $S^\star$ *sorted by non-decreasing f (ties by i)* and the ground-truth answer is $|S^\star|$. This yields a single canonical `\ids{...}` and `\answer{...}` for each instance.

Complexity. Preprocessing and $p(i)$ by binary search take $O(m \log m)$; the DP pass and backtracking are $O(m)$; greedy verification is $O(m)$.

A.2 Longest Increasing Subsequence (Unique Optimum)

Problem model. Given integers $a_1, \ldots, a_n$, an LIS is a subsequence $1 \leq i_1 < \cdots < i_L \leq n$ with $a_{i_1} < \cdots < a_{i_L}$. We allow duplicate values in (a_i); the strict inequality enforces strictly increasing